

ILDAR SHAYAKHMETOV

Senior / Staff Software Engineer | AI Products | Zero-to-One Engineering

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Summary

Hands-on senior/staff engineer with 10 years of experience building backend platforms and product surfaces across AI, fintech, analytics, telecom, and banking. Strongest when an idea is still rough: clarify the user problem, challenge unnecessary scope, choose an architecture that can ship, and stay with the feature through production. Deep in distributed JVM systems, with practical full-stack delivery in TypeScript, React, Next.js, Expo, and Telegram Mini Apps. Has led backend teams, reviewed product and technical decisions, mentored engineers, and built several independent products from zero.

Core Skills

- **Product ownership:** turn ambiguous ideas into user flows, technical decisions, working software, and operational feedback
- **AI product systems:** LLM orchestration, typed proposals, deterministic validation, capability boundaries, audit history, human approval
- **Full-stack delivery:** TypeScript, NestJS, React, Next.js, Expo, React Flow, REST APIs, authentication, scheduling, internationalization
- **Backend depth:** Java, Kotlin, Scala, Rust, Spring Boot, Project Reactor, Akka Streams, Kafka, RabbitMQ, PostgreSQL, Redis
- **Engineering leadership:** architecture direction, design and code reviews, prioritization, mentoring, hiring, cross-team alignment

Projects

HealthTracer - AI health companion platform

Turned a broad AI-coach idea into a structured product across mobile, web, and backend. Designed a multi-domain LLM pipeline where the model proposes typed changes, while the backend validates them and the user approves each immutable revision. Added a deterministic planner, explicit capability boundaries, and code-enforced safety floors so the product remains useful without giving the model unchecked authority. *TypeScript, NestJS, Next.js, Expo, PostgreSQL.*

bot-constructor - visual automation product

Built the product end to end: React Flow interaction model, four reactive Kotlin services, and an execution engine for 23 node types including triggers, branches, HTTP, sandboxed code, schedules, and AI/service connectors. Owned the less visible product work too - encrypted credentials, per-node history, failure visibility, and deployable Docker/Kubernetes packaging. *Kotlin, Spring WebFlux, RSocket, MongoDB, React.*

BeoSand - Telegram-first booking product

Built a player-facing Telegram Mini App and a React admin console for real booking operations in Belgrade. Kept pricing, availability, and capacity server-authoritative while making the user flow quick and understandable; added scheduling, booking requests, broadcasts, and RU/SR/EN localization. *TypeScript, React, Vite, Telegram Mini Apps.*

Professional Experience

Software Developer - Paysend

Feb 2023 - Jun 2026

Belgrade, Serbia

- Owned production payment services in Java and Kotlin, from implementation and external integrations through testing, delivery, and operational follow-up.
- Delivered PCI-DSS-compliant functionality and transaction-security controls for critical money-movement flows.
- Led a GraalVM native-image migration: average memory use fell from about 20 MB to under 2 MB, and allocated memory from 200 MB to 20 MB. The migration saved approximately EUR 20,000 per month in cloud costs.
- Improved asynchronous RabbitMQ interactions and GitLab CI/CD with JUnit and Testcontainers; contributed to hiring and backlog decisions.

Professional Experience - Continued

Team Lead - Prizma.tools

Nov 2021 - Jan 2023

- Led a small backend team building an analytics product, translating incomplete stakeholder ideas into concrete priorities, architecture, and shipped services.
- Stayed hands-on while setting direction for fault-tolerant services, data pipelines, storage, caching, and asynchronous processing.
- Introduced practical review, testing, monitoring, and refactoring habits; mentored engineers and spread ownership instead of centralizing every decision.

Software Engineer - MTS

Aug 2019 - Oct 2021

Moscow, Russia

- Designed a real-time platform processing more than 5 million Kafka messages per second in aggregate across the platform, where backpressure, latency, deduplication, and failure behavior were product constraints rather than afterthoughts.
- Built the core engine in Java and Akka Streams and consumer services in Scala; used Aerospike caching to protect high-volume data flows from duplicate work and avoidable latency.
- Mentored interns and new hires and ran internal Scala sessions to shorten the path from unfamiliar code to useful contribution.

Software Engineer - Raiffeisen Bank

May 2018 - Aug 2019

Moscow, Russia

- Built and evolved a high-load notification product in Java and Spring Integration, including a rewritten mass-notification flow and a new history service.
- Redesigned caching and added React/WebSocket functionality so users received faster, clearer feedback from backend events.

Earlier Engineering Roles - EPAM Systems, Trendsoft, Dixy

2016 - 2019

Delivered data-validation services and REST APIs, integrated business systems including 1C and AmoCRM, and developed Android applications with Retrofit and RxJava.

Selected Engineering Outcomes

- Shipped systems ranging from consumer mobile and web products to payment services and a streaming platform processing **5M+ messages/second in aggregate**.
- Reduced production infrastructure cost by approximately **EUR 20,000 per month** through a measured runtime migration.
- Built product logic where correctness matters: payment controls, server-authoritative pricing and capacity, typed AI proposals, and auditable decisions.
- Worked across the whole delivery loop: shaping requirements, architecture, implementation, tests, CI/CD, production diagnosis, and follow-up improvements.

Technical Skills

- **Languages:** Java, Kotlin, Scala, Rust, TypeScript, Python, SQL
- **Product & Web:** React, Next.js, Expo, React Flow, Telegram Mini Apps, REST, WebSockets, i18n
- **Backend & Data:** Spring Boot, Spring WebFlux, Project Reactor, Akka Streams, NestJS, Kafka, RabbitMQ, PostgreSQL, MongoDB, Aerospike, Redis
- **AI Systems:** LLM-backed workflows, multi-agent fan-out and synthesis, typed outputs, deterministic planners, human-in-the-loop approval
- **Delivery:** Docker, Kubernetes, GitLab CI/CD, GitHub Actions, GraalVM, JUnit, Testcontainers, Git

Education & Languages

HSE University - B.Sc. in Software Engineering, 2014 - 2018

English - Advanced (C1) | Russian - Native